

Nhlanhla Shakudyiwa

Full Stack Developer

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PROFILE

Computer Systems Engineering (ICT) recent graduate and Full Stack Developer with hands-on experience in software development, embedded systems, networking, and system integration.

Proficient in PHP (Symfony), C/C++, C#, Python, HTML, CSS, JavaScript, and web technologies, with experience using VS Code, Git, Bitbucket, Jira, Docker, web servers, Unity, ML-Agents, microcontrollers (ESP32, Arduino, PIC), and automation systems.

During my internship, I gained practical experience with enterprise platforms, software development and integrations, including Jira, iTop, SigningHub, and middleware solutions, strengthening skills in system integration, API usage, technical documentation, database management, environment management (DEV, QA, UAT, PROD), and collaborative problem-solving.

Passionate about applying technical knowledge to solve real-world problems and contribute to innovative software development.

EDUCATION

Computer Systems Engineering

Pretoria, South Africa

Tshwane University of Technology

- Software Development, C++, C#, and Python, Embedded Systems, Networking, Databases (MySQL), Operating Systems, PLC Programming, Digital Electronics, Logic Design, Artificial Intelligence, Mathematics and System Integration.
- Applied object-oriented design (OOP), algorithm development, and data structures to develop scalable and efficient programs.
- Worked on hardware-software integration projects using microcontrollers (ESP32, Arduino, PIC) and automation systems.
- Developed strong analytical and problem-solving skills through projects involving control systems, system modeling, and AI applications in Unity

LANGUAGES

English



Xitsonga



Isizulu



Venda



PROFESSIONAL WORK EXPERIENCE

Full Stack Developer

08/2025 – Present

MfundoPedia

Sandton, South Africa

SigningHub Integration – work Project

- Integrated SigningHub workflows with enterprise systems for document signing and tracking.
- Configured and tested workflow automations and user access processes.
- Applied API integration, environment management, and technical documentation to ensure smooth operations.

CRM Development – work Project

- Developed a Customer Relationship Management (CRM) system to manage client interactions and automate workflows.
- Designed databases, user interfaces, and business logic to streamline reporting and client management.
- Strengthened skills in backend development, database management, and system integration.

- Designed solution architecture documentation for Jira integrations, outlining system flow, data exchange, and environments.
- Designed technical documentation explaining SigningHub API authentication, endpoints, and usage reporting.
- Designed a documentation in comparison of better CRM between Vtiger and excalibur.
- Developed and maintained web applications using PHP (Symfony), HTML, CSS, JavaScript, and backend integrations.
- Worked with enterprise platforms including Jira, iTop, SigningHub, and middleware solutions to implement system integrations.
- Participated in requirements analysis and solution design discussions.
- Improved professional skills in technical documentation, system analysis, and integration workflows.
- Designed and tested RESTful API integrations using Postman and PHP scripts.
- Handled JSON request/response parsing, HTTP status codes, and error scenarios.
- Documented API contracts, request flows, and response structures for internal use.
- Gained experience in API usage, environment management (DEV, QA, UAT, PROD), database management, and technical documentation.
- Enhanced collaborative problem-solving skills through cross-team projects and integration workflows.

SKILLS

Technical Skill

- **Programming & Development:** C/C++, Python, PHP (Symfony), HTML, CSS, JavaScript
- **Database & Data Management:** SQL, ETL, Database Design
- **Software & Systems:** Object-Oriented Programming (OOP), Algorithm Design, Software Engineering, Operating Systems, IDEs (VS Code, Embarcadero, Unity)
- **Embedded & Digital Systems:** Microcontroller Programming (ESP32, Arduino, PIC), Digital Electronics, PLC Programming
- **Mathematics & Engineering Concepts:** Matrices, Vectors, Linear Algebra, System Modeling
- **Tools :** VS Code, Git, Bitbucket, Docker, Jira, Teams, Linux (Ubuntu)

Soft Skills

- Clear communication (verbal and written)
- Kind, friendly, and approachable
- Hardworking, dedicated, and self-motivated
- Leadership and team motivation
- Team player with collaborative mindset
- Quick learner and adaptable to new technologies

PROJECTS

Secure Access Dashboard with Biometric Authentication

09/2025 – 11/2025

- **Secure Access Dashboard with Biometric Authentication** – University Project
Designed and implemented a secure access system with biometric authentication for a dashboard interface. Managed user authentication, data protection, and access control using modern security practices.
- **Technologies used:** PHP, MySQL, Biometric Sensors, Web Dashboard
- **Skills gained:** Security protocols, Database handling, Front-end development

Automated Solar Tracking System– University Project

05/2025 – 05/2025

- Built a solar tracking system using Arduino Uno, sensors, and motors to optimize solar panel positioning. Implemented data logging for performance tracking and system monitoring.
- **Technologies used:** Arduino C++, Sensors, Motors, Embedded Systems
- **Skills gained:** Microcontroller programming, Embedded systems, IoT

Banking Application – Embarcadero (File-Based Storage)

10/2024 – 10/2024

- Developed a student banking application featuring secure user registration and local file-based storage. Managed smooth financial transactions and implemented secure access for multiple users.
- **Technologies used:** Embarcadero C++, File-Based Storage, Database Management

- **Skills gained:** Software development, Team leadership, Testing and debugging, Database handling

K-Means Clustering Visualization in Unity – University Project

08/2024 – 08/2024

- Implemented **k-means clustering** for dataset analysis and visualized results using Unity Gizmos. Applied data normalization and statistical analysis to display meaningful patterns.
- **Technologies used:** python, pytorch, tensorflow, Unity, C#, Data Clustering, Visualization
- **Skills gained:** Data analysis, 3D visualization, Unity development

CERTIFICATES

AWS Certification

- AWS Cloud Practitioner Essentials

KWENA MINING &

METALLURGICAL SERVICES

New development programme

REFERENCES

Available Upon Request